

Linear Regression Analysis
Automobile Attributes and Miles Per Gallon

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ALY 6020: Predictive Analytics

Designing for Fuel Efficiency: A Linear Regression Analysis

A car manufacturer known for producing large automobiles faces declining sales and has commissioned this analysis to identify which vehicle design attributes most strongly predict fuel efficiency. Using a dataset of 398 vehicles from model years 1970 to 1982, a linear regression model was built to predict miles per gallon (MPG) from seven attributes: cylinders, displacement, horsepower, weight, acceleration, model year, and origin. The analysis covers data cleansing, full model construction, and optimization via backward elimination, concluding with concrete design recommendations.

Part 1: Data Cleansing and Preparation

Dataset Overview

The dataset contains 398 rows and 8 columns. The target variable is MPG, representing fuel efficiency. The seven predictor variables are Cylinders, Displacement (cubic inches), Horsepower, Weight (pounds), Acceleration (seconds to reach 60 mph), Model Year (two-digit, 70 to 82), and US Made (binary: 1 for US-manufactured, 0 for foreign). Upon initial loading, the dataset displayed no null values; however, a critical data quality issue was immediately apparent in the Horsepower column, which was read as a string data type rather than numeric. This anomaly indicated the presence of non-numeric placeholder values embedded within the column.

Missing Value Detection and Imputation

The Horsepower column contained six rows with question mark placeholders for missing values, a common convention in legacy datasets. These were replaced with NaN using `pd.to_numeric()` with `errors='coerce'`, converting the column to float64 and exposing the missing values cleanly. Since six records represent only 1.5% of the dataset, imputation was preferred over row deletion to preserve the full sample. The median value of 93.5 horsepower was selected over the mean because the Horsepower distribution exhibits a rightward skew driven by high-performance outliers, making the median a more robust choice. After imputation, the dataset contained zero missing values across all columns.

Duplicate and Range Validation

A duplicate check returned zero fully duplicated rows. Descriptive statistics confirmed all variable ranges were plausible for 1970s and 1980s vehicles: MPG ranged from 9.0 to 46.6, Weight from 1,613 to 5,140 pounds, and Cylinders included values of 3, 4, 5, 6, and 8, all valid production configurations.

Outlier Assessment

Outlier detection was performed using the Interquartile Range (IQR) method, where the lower and upper bounds were defined as Q1 minus 1.5 times the IQR and Q3 plus 1.5 times the IQR respectively. Results were summarized

in a table displaying Q1, Q3, IQR, bounds, and outlier count for each column, with boxplots generated for visual confirmation. Flagged records, primarily high-MPG compact vehicles from the late 1970s oil crisis era, were retained rather than removed as they represent legitimate real-world observations. Removing them would distort the model by eliminating the most fuel-efficient vehicles in the dataset.

Distribution Analysis

Histograms revealed right skew in MPG, Horsepower, Displacement, and Weight, and a bimodal pattern in Cylinders concentrated at four and eight, reflecting the mix of foreign compacts and domestic V8s. These distributional characteristics were noted for residual diagnostics but did not require transformation for OLS regression. (Appendix A)

Part 2: Linear Regression Modeling

Correlation Analysis

A correlation heatmap revealed that Weight had the strongest negative correlation with MPG at -0.83, followed by Displacement (-0.80), Cylinders (-0.78), and Horsepower (-0.77). Model Year showed a positive correlation of +0.58, reflecting cumulative engineering improvements. US Made correlated at -0.57, indicating foreign vehicles in this dataset achieved higher MPG. Critically, the engine-related predictors were also strongly intercorrelated, flagging potential multicollinearity. (Appendix B)

Model Specification

MPG was specified as the dependent variable. All seven remaining columns served as predictors. The data was split 80/20 into training (318 records) and test sets (80 records) with random_state=42. A scikit-learn LinearRegression model was fitted on the training set, with p-values computed manually via the t-distribution from residual variance and the design matrix.

Feature	Coefficient	Std Error	T-Statistic	P-Value
Intercept	-18.7403	—	—	—
Weight	-0.0070	0.0007	-9.604	< .001
Model Year	0.8216	0.0602	13.654	< .001
US Made	-2.8045	0.5559	-5.045	< .001
Displacement	0.0197	0.0089	2.217	.027
Horsepower	-0.0152	0.0155	-0.982	.327
Acceleration	0.0675	0.1096	0.616	.539
Cylinders	-0.1762	0.3771	-0.467	.641

Model Performance.

The full model achieved a test R-squared of 0.8463 and adjusted R-squared of 0.8313, explaining approximately 84.6% of MPG variance. RMSE was 2.875 MPG and MAE was 2.262 MPG, indicating predictions deviated by roughly 2.3 MPG on average. These metrics reflect strong linear model performance.

Significant Predictors.

Weight ($p < .001$, $t = -9.604$), Model Year ($p < .001$, $t = 13.654$), and US Made ($p < .001$, $t = -5.045$) were statistically significant. Displacement was marginally significant ($p = .027$). Cylinders, Horsepower, and Acceleration were not significant, attributable to severe multicollinearity with Weight rather than a true absence of effect. The residual mean was 0.081 with a standard deviation of 2.892, and residual diagnostics showed approximately random scatter around zero, confirming that OLS assumptions were reasonably met. (Appendix C and D).

The 95% confidence intervals for the three significant predictors further confirms their reliability. The interval for Weight was $[-0.008426, -0.005560]$, entirely below zero, confirming that heavier vehicles consistently achieve lower MPG. The interval for Model Year was $[0.703212, 0.940017]$, entirely above zero, confirming consistent year-on-year efficiency gains. The interval for US Made was $[-3.898330, -1.710684]$, entirely below zero, confirming the domestic MPG gap. None of these intervals contain zero, which is consistent with their p-values below 0.001.

Part 3: Model Optimization and Design Recommendations

Multicollinearity via VIF

Variance Inflation Factors confirmed severe multicollinearity: Weight (VIF = 131.5), Cylinders (119.5), Model Year (109.6), Displacement (101.1), Acceleration (67.3), and Horsepower (61.6) all far exceeded the threshold of 10. Only US Made was acceptable at 5.2. These results confirm that the engine-related variables are statistically redundant in the presence of each other.

The extreme magnitude of these VIF values, several exceeding 100, arises because Cylinders, Displacement, Horsepower, and Weight are not independent physical quantities. They are near-linear combinations of the same underlying construct: engine size. A vehicle with eight cylinders will almost always have high displacement, high horsepower, and high weight. These variables move together so predictably across the dataset that regressing any one of them on the others yields an R-squared close to 1.0, which is precisely what a VIF above 100 indicates. This does not mean the predictors are unimportant; it means their individual effects cannot be separated within a single OLS model. Backward elimination resolves this by retaining only the predictor (Weight) that most efficiently captures the shared variance, without the redundancy penalty.

Backward Elimination

Starting with all seven predictors and iteratively removing the feature with the highest p-value, backward elimination removed Cylinders ($p = .641$), Acceleration ($p = .522$), Horsepower ($p = .091$), and Displacement ($p = .059$). The three retained predictors, Weight, Model Year, and US Made, were all significant at $p < .001$.

Metric	Full Model (7 predictors)	Optimized Model (3 predictors)
R-Squared	0.8463	0.8388
Adjusted R-Squared	0.8313	0.8324
RMSE	2.8749	2.9441
MAE	2.2620	2.2943

Model Evaluation.

The optimized model preserved nearly identical predictive accuracy with four fewer predictors. The adjusted R-squared improved slightly from 0.8313 to 0.8324, confirming the removed variables were not contributing independently. RMSE and MAE increased only marginally (by 0.07 and 0.03 respectively), a negligible practical difference. The model can achieve the manufacturer's goals: it explains over 83% of MPG variance using three measurable design attributes, enabling quantitative MPG prediction for proposed vehicle configurations before prototyping. (Appendix E)

Design Recommendations.

The following coefficients are drawn from the optimized three-predictor model and differ slightly from the full model values reported in Part 2, as refitting on three features redistributes the explained variance. Weight reduction is the highest-priority action. The coefficient of -0.0060 means each pound removed improves predicted MPG by 0.006. A 500-pound reduction yields approximately 3 additional MPG. The manufacturer should adopt lightweight materials such as aluminum body panels, high-strength steel alloys, and composite structural components.

The Model Year coefficient of +0.8070 reflects cumulative gains from fuel injection, improved transmission efficiency, and aerodynamic optimization. Investing in these engineering improvements is the second key lever. Although Cylinders, Displacement, and Horsepower were eliminated by backward elimination due to multicollinearity, their effects are captured through weight: reducing vehicle mass implicitly necessitates smaller, lower-displacement engines with fewer cylinders. The US Made coefficient of -2.158 indicates US vehicles achieved approximately 2.2 fewer MPG than foreign equivalents in this period, a baseline gap the manufacturer can close primarily through weight reduction and engineering modernization.

To demonstrate the model's practical utility, consider two vehicle configurations. A typical 1970 US vehicle weighing 4,000 pounds produces a predicted MPG of 11.70. A redesigned 1982-equivalent vehicle weighing 2,800 pounds, built to foreign engineering standards, produces a predicted MPG of 30.70, an improvement of 19.00 MPG. This comparison illustrates concretely how the three levers identified by the model, weight reduction, engineering modernization, and platform redesign, combining to close the fuel efficiency gap. The manufacturer can apply this equation to any proposed configuration before committing to a prototype, making the model a practical design tool rather than a retrospective analysis.

Conclusion

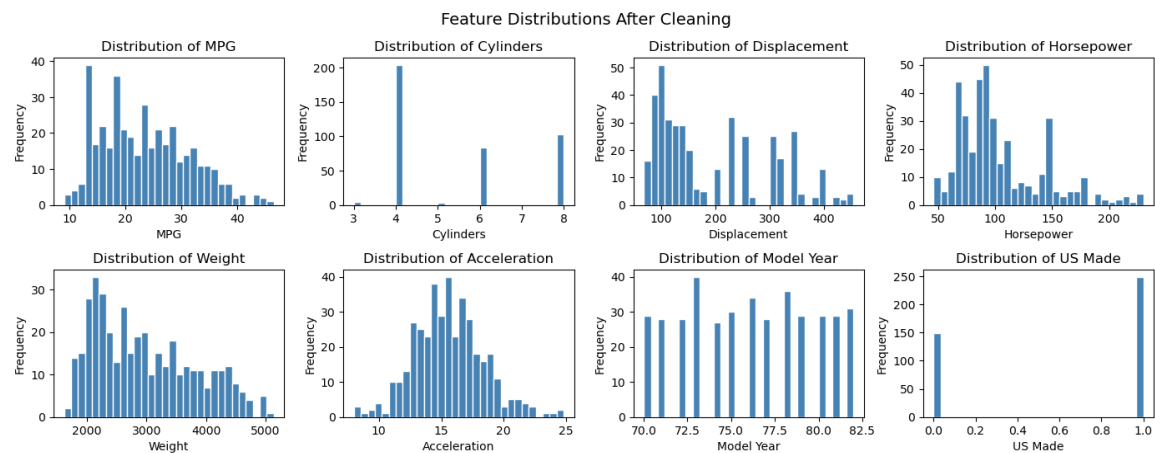
This analysis built and optimized a linear regression model to identify the vehicle design attributes most predictive of fuel efficiency. After thorough data cleansing, the full seven-predictor model explained 84.6% of MPG variance on the test set. Backward elimination produced a lean three-predictor model with an improved adjusted R-squared of 0.8324, retaining only Weight, Model Year, and US Made. Vehicle weight was the dominant predictor, and its reduction is the single most impactful design decision available to the manufacturer. Combined with modern engineering investments and lighter platform architecture, the regression model provides a clear, quantitative roadmap for designing competitive, fuel-efficient automobiles.

References

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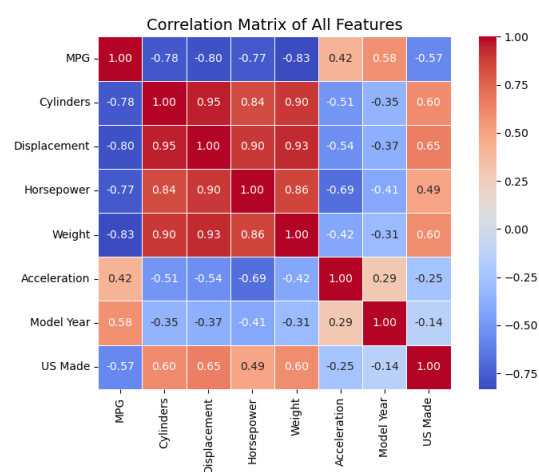
Appendix: A

Feature Distributions After Cleaning.



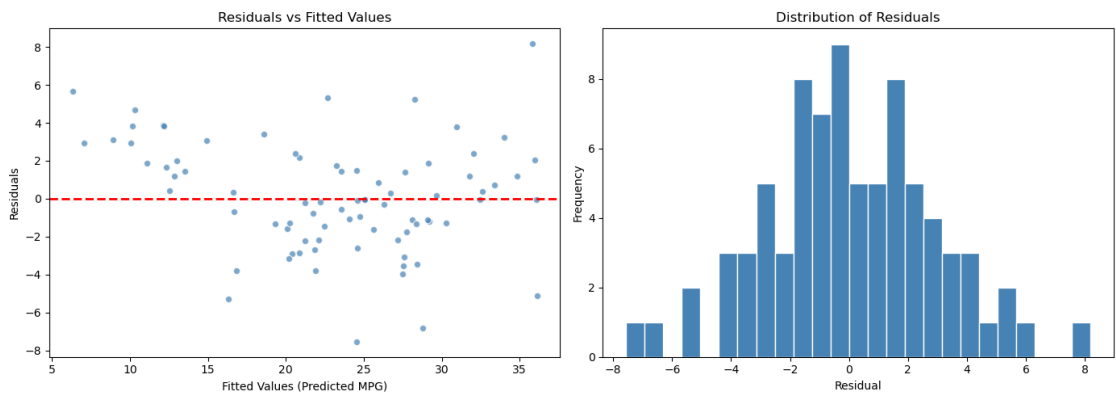
Appendix: B

Correlation Matrix of All Features

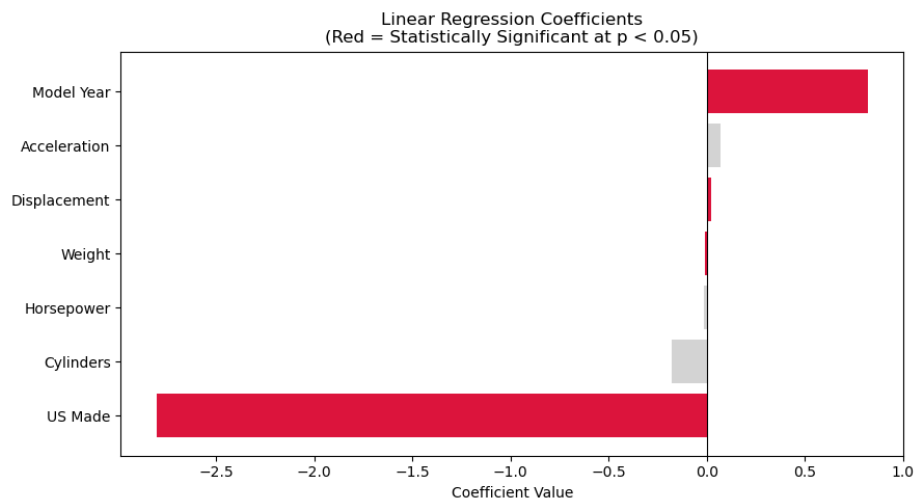


Appendix: C

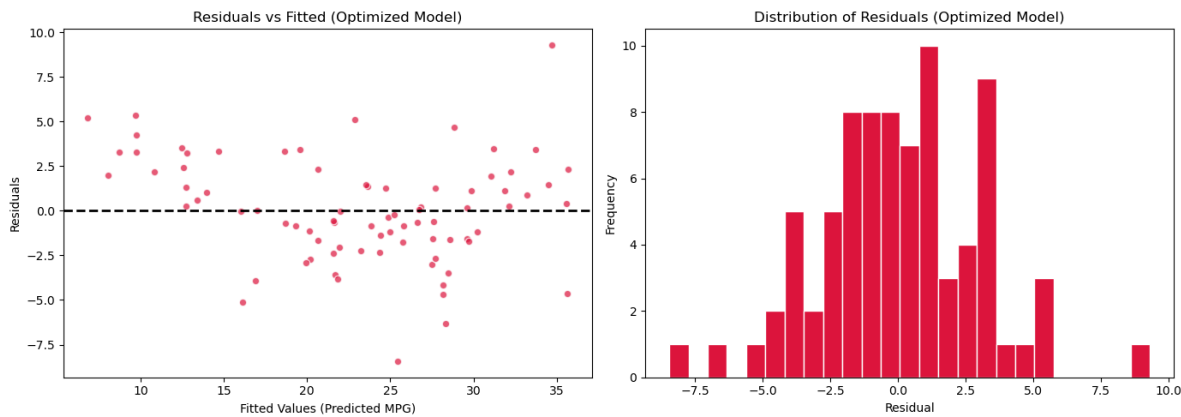
Residuals vs. Fitted Values and Distribution of Residuals (Full Model)



Appendix: D
Linear Regression Coefficients by Significance (Full Model)



Appendix: E
Residuals vs. Fitted Values and Distribution of Residuals (Optimized Model)



Appendix: F
Q-Q Plot of Residuals

